

Breaking the Three Bottlenecks That Stop Faculty From Adopting Mastery Grading

Redesigning AAE 33700, a required 200+-student propulsion course, with Claude

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Primary AI Tool Used: Claude (Anthropic)

Why faculty don't adopt standards-based grading

- **Problem Description:** The pedagogy is well-evidenced. Adoption stalls on **implementation cost** — not on doubt about whether it works.
 - initial faculty workload
 - fitting standards into a points-based LMS
 - score consistency across graders.
- **Impact Goal:** Provide a ready to go implementation of standards based grading for AAE 33700 along with the Claude Skill file for others
- **Course Context:** AAE 33700, 200 – 300 students per semester; West Lafayette
- **Current Limitations:**
 - No Paybook: standards-based grading in engineering courses has not been done in AAE
 - **Scale makes it worse:** every reassessment needs a fresh, equivalent problem. $17 \text{ standards} \times 3 \text{ attempts/student}$ with different problems for different weeks means hundreds of problems are required.

Three bottlenecks, three ways I used Claude

- I treated Claude as a design partner and wrote the recurring judgment calls into a **reusable method**
 - **1 · Structure & logistics** — interactive dialogue across seven design phases. At each fork Claude named the tradeoff; I decided.
 - **2 · Standards from artifacts** — Claude read my syllabus, lecture slides, and past exams, then drafted “I can...” statements with scope notes.
 - **3 · Sustainable reassessment** — templated, de-scaffolded problem versions per standard, with every answer computed in code.
 - **The guardrails that made it trustworthy:** anchor difficulty to a real past exam; confirm convention before computing; reconcile answers to the students’ own tables; send no student data.

Barrier → what Claude did → what came out

- *Same three bottlenecks the literature names — attacked at three different points in the redesign.*

1 • STRUCTURE & LOGISTICS

“I believe in this, but I don’t know how to build it — or make my gradebook do it.”

WHAT CLAUDE DID

Interactive dialogue across seven design phases. At each fork it named the tradeoff; I decided.

WHAT CAME OUT

17 standards in 5 modules, bundled ladder + ethics gate, reassessment calendar, Brightspace status board.

2 • STANDARDS FROM ARTIFACTS

“Writing 17 clean, assessable standards from scratch is weeks of work.”

WHAT CLAUDE DID

Read my syllabus, lecture slides, and past exams; drafted “I can…” statements with scope notes.

WHAT CAME OUT

Student-facing standards plus private scope notes that tell graders what counts as met.

3 • SUSTAINABLE REASSESSMENT

“Every retake needs a fresh problem. At 300 students that’s hundreds.”

WHAT CLAUDE DID

Built interchangeable versions from a per-standard template — de-scaffolded, difficulty anchored to prior exams.

WHAT CAME OUT

Verified banks per standard — every answer computed in code, reconciled to the course tables.

Every column took several passes. First drafts were wrong — the value was in the iteration and verification, not the speed.

What got built — including what broke first

- **Built:** 17 standards across 5 modules, a bundled letter-grade ladder with an ethics pass-gate, a 3-level proficiency scale, a reassessment calendar, and a Brightspace status-board gradebook.
- **Problem banks:** multi-version interchangeable problems per standard — every answer computed in code and reconciled to the students' own reference tables, not estimated by the model.
- **Accessible by design:** ADA Title II / WCAG 2.1 AA throughout — real Office Math equations, proper heading hierarchy, AA-contrast color.
- **What failed first:** one bank was framed from assumed product moles instead of reactant feeds; another shipped six near-identical versions; difficulty came in too low more than once. Each was caught on review and rebuilt — twice, in one case.
- **The deliverable:** the whole method packaged as a portable Claude Skill — seven phases plus five reference files any instructor can load.

What I learned & what's next

LESSON LEARNED

- It's iterative — not one prompt and done. The quality came from refining across many passes.

REPLICATE IT

- Load the Claude Skill, then give it your syllabus, lecture slides, and assessments.

WHAT'S NEXT

- Deploying in AAE 33700 in Fall 2026 — stay tuned to find out how it goes.